FATURA ANALIZ

Düzce University
Faculty of Engineering

PRESENTATION 05 — FINAL DEFENSE

An AI-Supported Expense & Invoice *Analysis System*

Undergraduate thesis defense — turning receipts into categorized expenses with Google Gemini 2.5 Flash.

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Course
CE498 · Project Design

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AGENDA

01–02

Problem & aims

Motivation, the gap in prior work, and four research questions.

03–04

System & AI

Architecture, the product, the Gemini engine and security.

05–06

Results & outlook

Findings, RQ answers, contributions, limitations & future work.

01 — MOTIVATION

Consistent expense tracking fails because *manual entry* is too much friction — receipts pile up, photos get lost, payments scatter.

Existing apps rely on rigid rules, read only text, and bury their AI inside the backend — leaving photographed invoices unprocessed.

01 — AIM & RESEARCH QUESTIONS

Turn the user into a *reviewer,* not a typist.

Submit an invoice image; receive an auto-generated, categorized expense in return.

RQ1	Accurate extraction from invoice images?
RQ2	Does AI categorization cut manual effort?
RQ3	Microservice without performance cost?
RQ4	Coherent end-to-end experience?

02 — WHERE THIS SITS IN THE LITERATURE

MANUAL

Lossy

Paper & spreadsheets.

OCR

Text capture

Rule-based categories.

ML

Classifiers

SVM, weak supervision.

LLM

Few-shot text

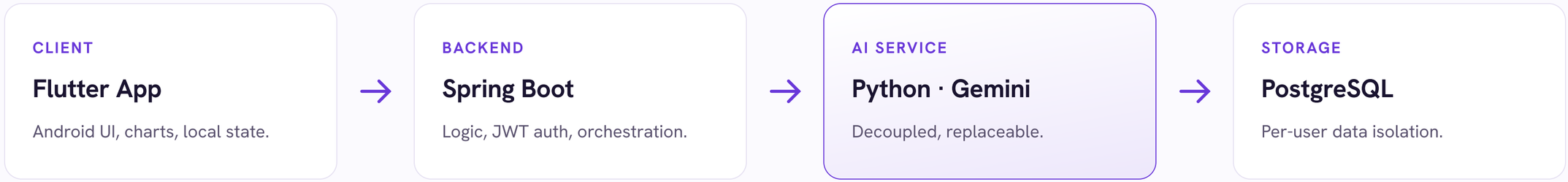
Little labeled data.

THIS WORK

Multimodal

Image + text, decoupled service.

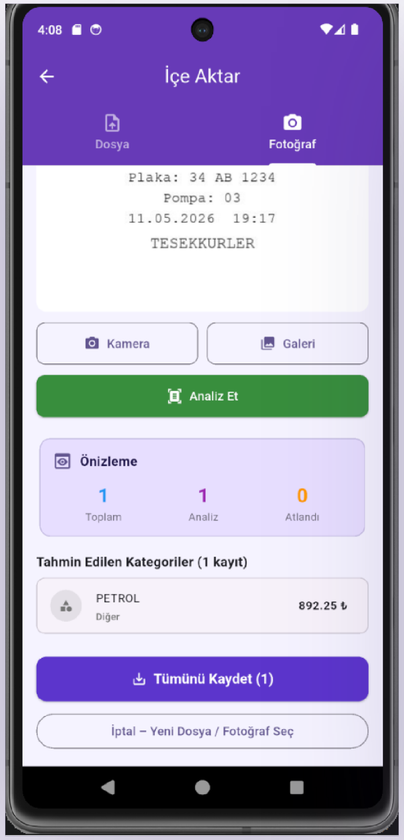
> One holistic inference replaces the OCR-then-classify pipeline — served from a replaceable microservice.



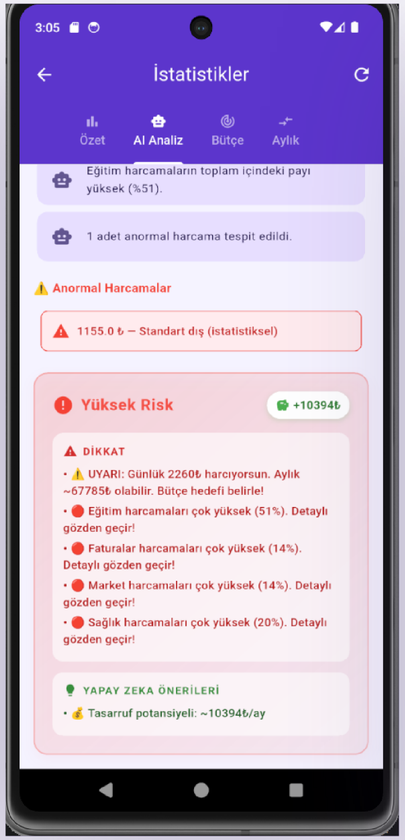
> Layered, loosely-coupled — the AI module can be upgraded without touching the backend.



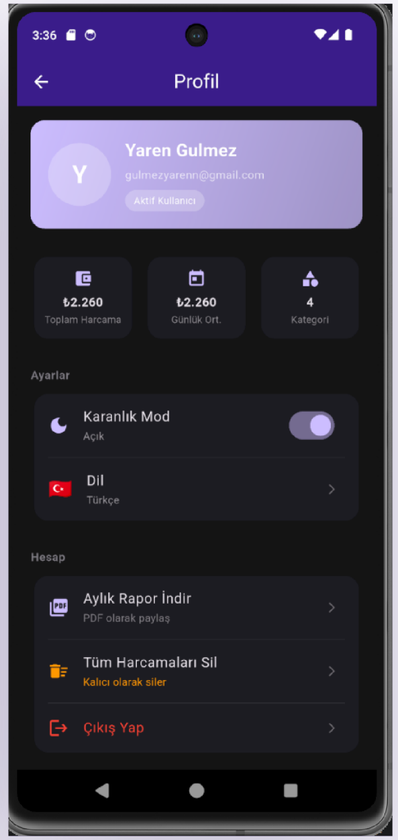
Home dashboard



AI invoice import



AI analysis & risk



Profile & dark mode

Image analysis

Extracts total & predicts category from a receipt photo — one multimodal call, no separate OCR stage.

Batch import

Classifies each CSV/XLSX row into one of ten categories from its description.

Commentary

Generates natural-language insight, anomaly flags, risk & saving estimates.

> Zero/few-shot prompting — no domain-specific training dataset required.

BCrypt

Salted password hashing.

JWT

Stateless, per-request validation.

Email verify

Pre-activation gate.

OTP reset

15-minute expiry.

> User-based isolation at both service & database layers — no cross-user access through any endpoint.

892₺

Petrol-receipt total extracted correctly

85%

CSV category accuracy (17/20)

100%

Backend & security tests passed

E2E✓

Full scenario validated, no errors

All defined functional & non-functional requirements were met.

RQ1 Extraction — **Confirmed across real receipts of varied layout.**

RQ2 Categorization — **85% accuracy clears the literature's satisfaction threshold.**

RQ3 Architecture — **Decoupled microservice added no measurable degradation.**

RQ4 End-to-end — **Seamless integration, consistent data across all screens.**

CONTRIBUTIONS

01 A fully operational AI-assisted platform

Goes beyond systems that merely *plan* AI integration — delivering working multimodal invoice analysis end-to-end.

02 LLM in place of a trained classifier

Strong categorization without large labeled datasets — more flexible and scalable than domain-specific models.

03 A maintainable, modular architecture

The AI layer can be upgraded, replaced or scaled independently of backend and client.

External dependency

Relies on the Gemini API — connectivity, latency & usage-based pricing; no offline model.

Cold-start anomalies

The 2σ anomaly rule needs enough history to be representative for new users.

Single-platform test

Validated on one Android device & account — no concurrent-load or iOS testing.

06 — FUTURE WORK

Income tracking

Net balance & savings-rate insight.

Multi-currency

Real-time conversion & aggregation.

Bank API

Auto-import transactions, no typing.

Fine-tuned model

Offline, lower-cost categorization.

> A structured user study would measure real-world usability & behavior change.

CONCLUSION

A functional, secure and AI-supported expense system — proving that LLM APIs like *Gemini 2.5 Flash* integrate cleanly into mobile finance through a lightweight microservice.

Manual entry, AI bulk import, automated receipt analysis, budgets and monthly commentary — the full personal-expense lifecycle in one app.

FINAL DEFENSE — THANK YOU

Thank you.
I welcome your questions.

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